



General Chemistry I

Recitation 9

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Nov. 24th

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
															Pnictogens	Chalcogens	Halogens			
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight C Solid Hg Liquid H Gas Rf Unknown Metals Alkali metals Alkaline earth metals Lanthanoids Actinoids Transition metals Post-transition metals Metalloids Nonmetals Reactive nonmetals Noble gases																2 He Helium 4.0026		
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122													5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305													13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798		
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29		
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71 89–103	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)		
7	87 Fr Francium (223)	88 Ra Radium (226)		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)		
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																				
			6 7	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97		
				89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)		

Quiz

1. Write simple valence bond wave functions for formation of bonds between C and H atoms. What C-H compound does the VB model predict? What geometry does it predict for the molecules?
2. Draw a Lewis electron dot diagram for PH_3 . Formulate the hybridization for the central atom of PH_3 and give the molecular geometry.

MO and VB theory

LCAO

- Ignores e-e repulsion
- Electrons wander around
- Orbitals: delocalized (离域)

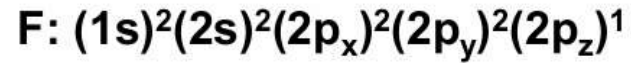
VB

- Exaggerates e-e repulsion
- Electrons adhere to the nuclei
- Orbitals: localized (定域/局域)

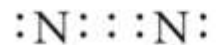
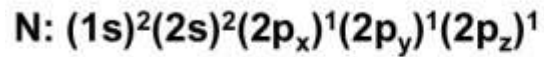
Grew out of the qualitative Lewis electron-pair model in which the chemical bond is described as a pair of electrons that is localized between two atoms.

- The VB model constructs a wave function for the bond by assuming that each atom arrives with at least one unpaired electron in an AO.
- The resulting wave function: a product of two one-electron wave functions, each describing an electron localized on one of the arriving atoms. (The spins of the electrons must be paired to satisfy the Pauli exclusion principle).

MO and VB theory

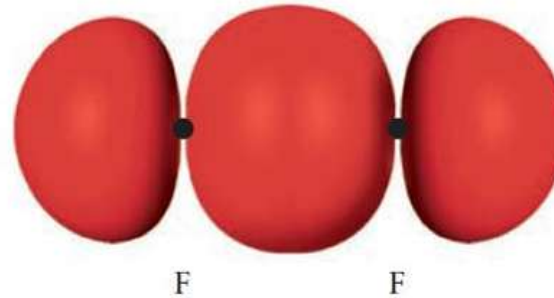


Lewis diagram for F₂.



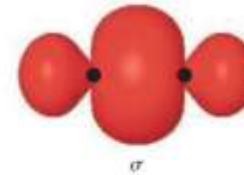
Lewis diagram for N₂.

$$\psi_g^{\text{bond}} = C_1[2p_z^A(1)2p_z^B(2) + 2p_z^A(2)2p_z^B(1)]$$



- When approach one another along the z-axis (2pz: head-on)

$$\psi_{\sigma}^{\text{bond}} = C_1[2p_z^A(1)2p_z^B(2) + 2p_z^A(2)2p_z^B(1)]$$



- The 2px and 2py orbitals: side-to-side.

$$\psi_{\pi_x}^{\text{bond}}(1, 2; R_{AB}) = C_1(R_{AB})[2p_x^A(1)2p_x^B(2)] + C_1(R_{AB})[2p_x^A(2)2p_x^B(1)]$$

$$\psi_{\pi_y}^{\text{bond}}(1, 2) = C_1[2p_y^A(1)2p_y^B(2) + 2p_y^A(2)2p_y^B(1)]$$



What about p_x , p_y ?

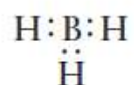
p_x , p_y or p_z ?

Please check carefully!

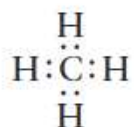
MO and VB theory



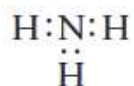
Beryllium hydride, BeH_2 , has four valence electrons, two from Be and one each from the two H atoms, all of which appear in the Lewis diagram. In VSEPR theory, the steric number is 2, so the molecule is predicted to be linear, and this prediction is verified by experiment. The electron configuration of the central atom is Be: $(1s)^2(2s)^2$. There are no unpaired electrons to overlap with H(1s) orbitals, so the VB model fails to predict the formation of BeH_2 .



Boron hydride, BH_3 , has six valence electrons corresponding to steric number 3 and a trigonal planar structure. With the electron configuration B: $(1s)^2(2s)^2(2p)^1$ on the central atom, the VB model cannot account for the formation of BH_3 and, in fact, predicts that BH is the stable molecule, which does not agree with experiment.



Methane, CH_4 , has steric number 4, and VSEPR predicts a tetrahedral structure, which is confirmed by experiment. Starting with the electron configuration C: $(1s)^2(2s)^2(2p)^2$, the VB model cannot account for the formation of CH_4 and predicts that CH_2 would be the stable hydride, which is again contrary to the experimental results.



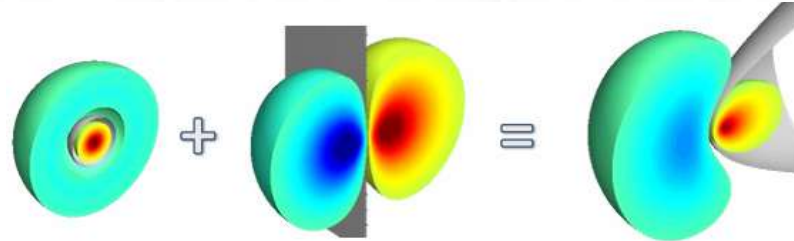
Ammonia, NH_3 , has steric number 4 with three shared pairs and one unshared pair on the N atom. VSEPR predicts a trigonal pyramid structure, as a subcase of tetrahedral structure, with angles slightly less than 109.5° due to repulsion between the unshared pair and the three bonding pairs. Experiment verifies this structure with angles of 107° . The electron configuration N: $(1s)^2(2s)^2(2p_x)^1(2p_y)^1(2p_z)^1$ would permit the formation of three σ bonds by overlap of H(1s) orbitals with each of the 2p orbitals on N. Because these 2p orbitals are all mutually perpendicular, VB predicts a trigonal pyramid but one with bond angles of 90° .

Hybridization for MO

BeH_2

$$\begin{vmatrix} \varphi_1 \\ \varphi_2 \end{vmatrix} = \begin{vmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{vmatrix} \begin{vmatrix} s \\ p_z \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{vmatrix} s \\ p_z \end{vmatrix}$$

$Be: (1s)^2(\chi_1)^1(\chi_2)^1$



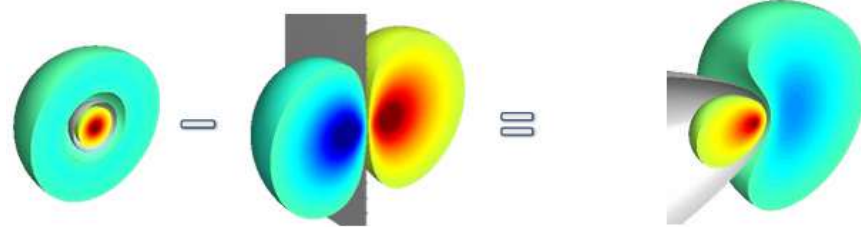
$$\chi_1(r) = \frac{1}{\sqrt{2}} [2s + 2p_z]$$

$$\frac{1}{\sqrt{2}} s$$

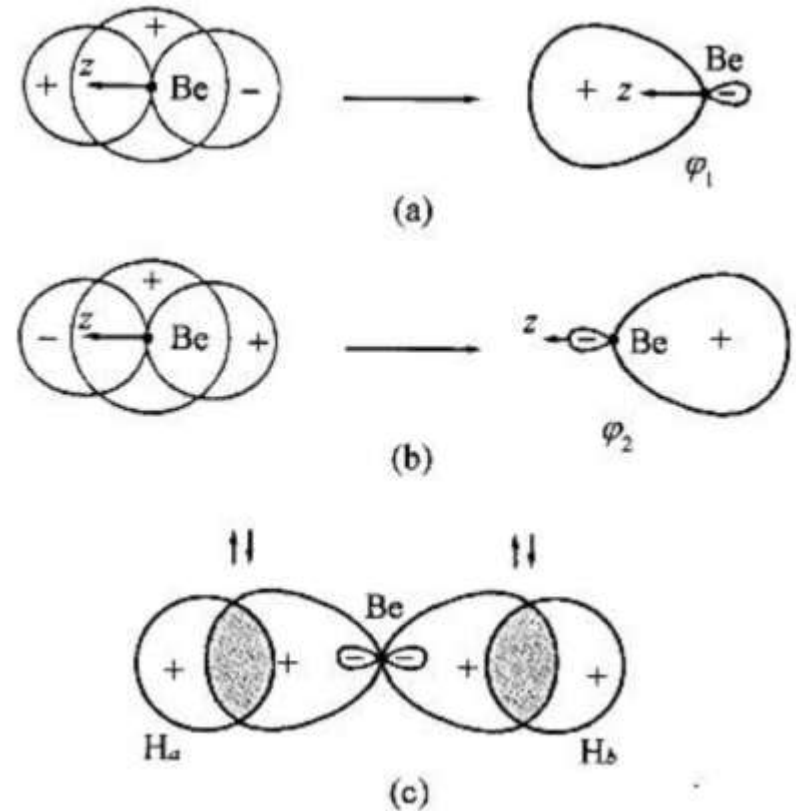
$$\frac{1}{\sqrt{2}} p_z$$

sp hybrid

$$\chi_2(r) = \frac{1}{\sqrt{2}} [2s - 2p_z]$$



Normalization constant



Hybridization for MO

CO₂ (sp) sp hybrid atomic orbitals: formed as the sum or difference of one s orbital and one p orbital.

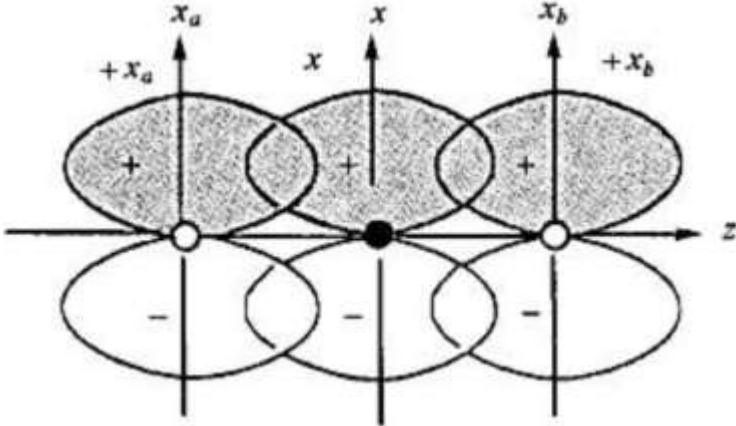
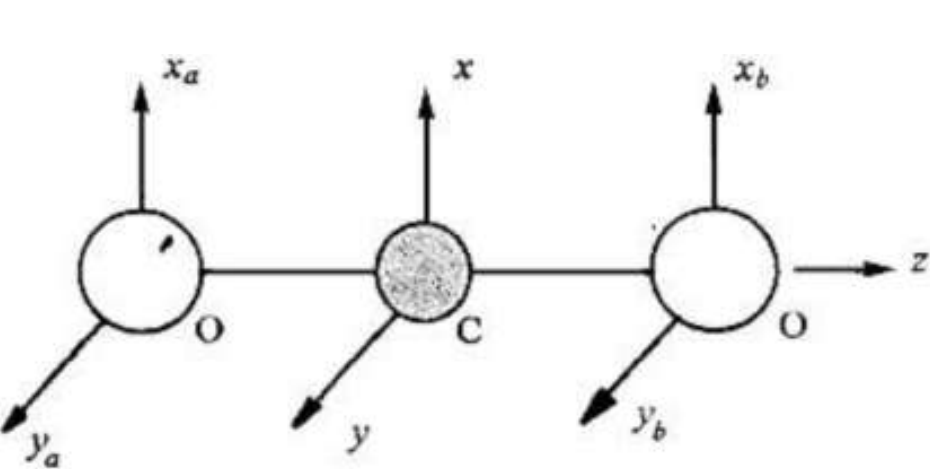
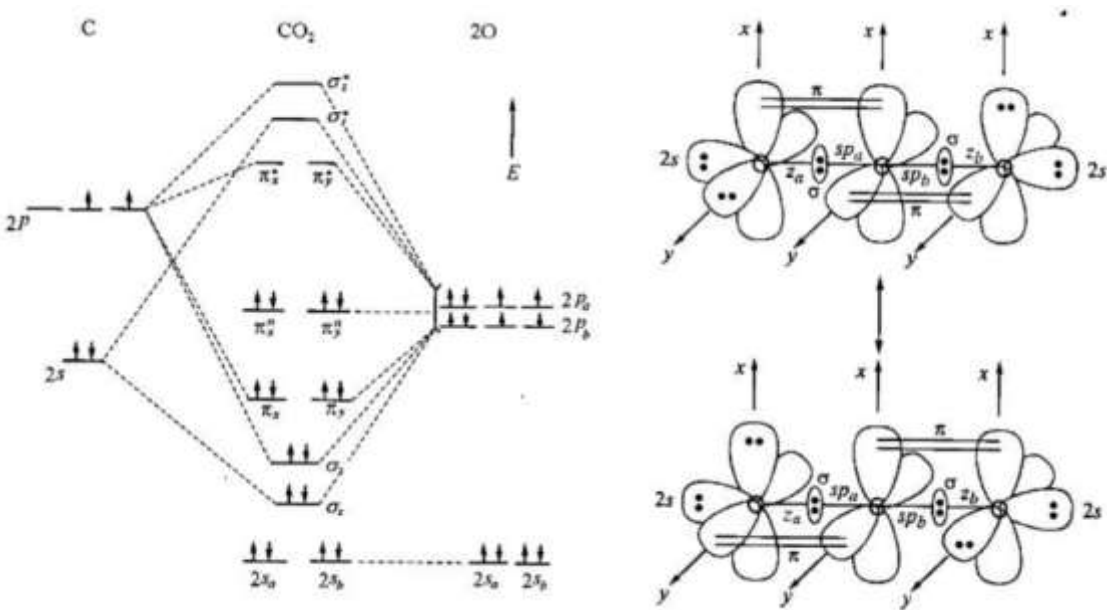


表 3.4.2 CO₂ 分子中分子轨道构建情况

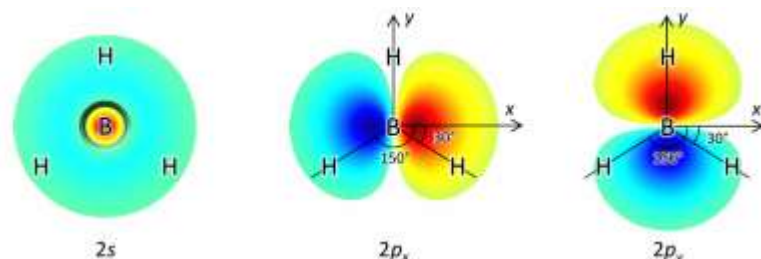
C 的轨道	O _a 和 O _b 的轨道	分子轨道
2s	$(1/\sqrt{2})[2p_x(a) + 2p_x(b)]$	σ_s, σ_s^*
2p _x	$(1/\sqrt{2})[2p_x(a) - 2p_x(b)]$	σ_x, σ_x^*
2p _x	$(1/\sqrt{2})[2p_x(a) + 2p_x(b)]$	π_x, π_x^*
2p _y	$(1/\sqrt{2})[2p_y(a) + 2p_y(b)]$	π_y, π_y^*
—	$(1/\sqrt{2})[2p_x(a) - 2p_x(b)]$	π_x^*
—	$(1/\sqrt{2})[2p_y(a) - 2p_y(b)]$	π_y^*



Hybridization for MO

BH₃ (sp²)

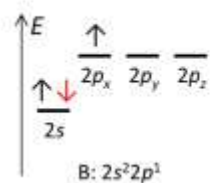
sp² hybrid atomic orbitals: formed from one s orbital and two p orbitals



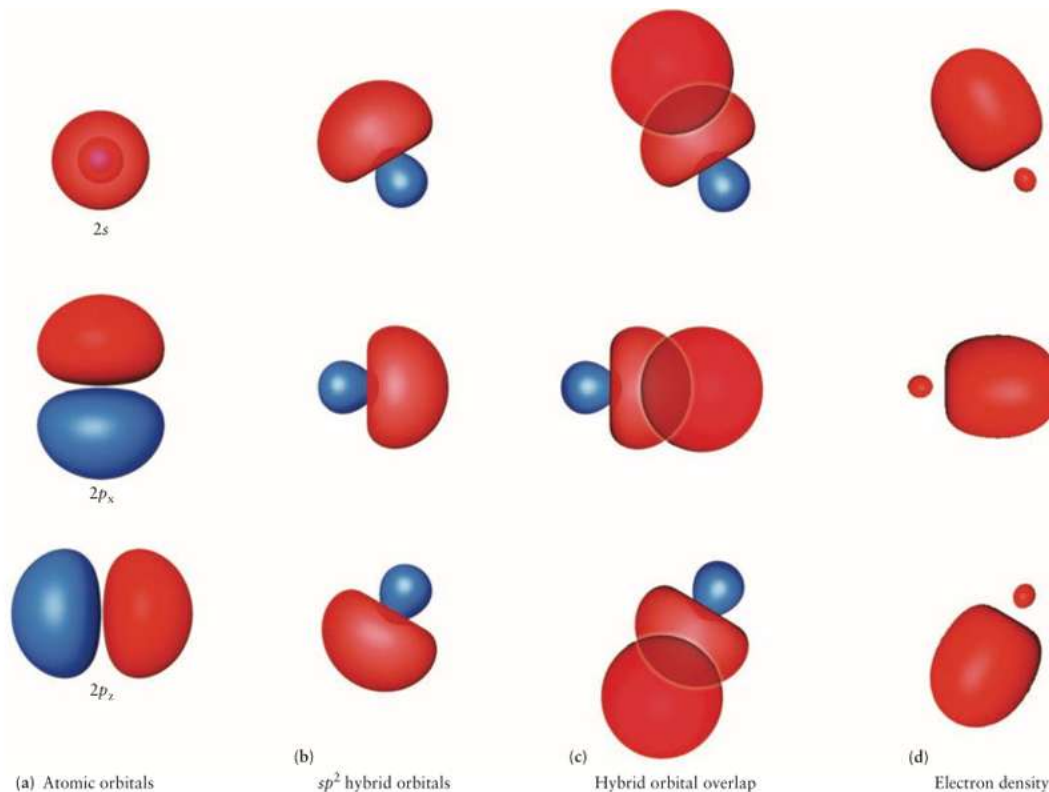
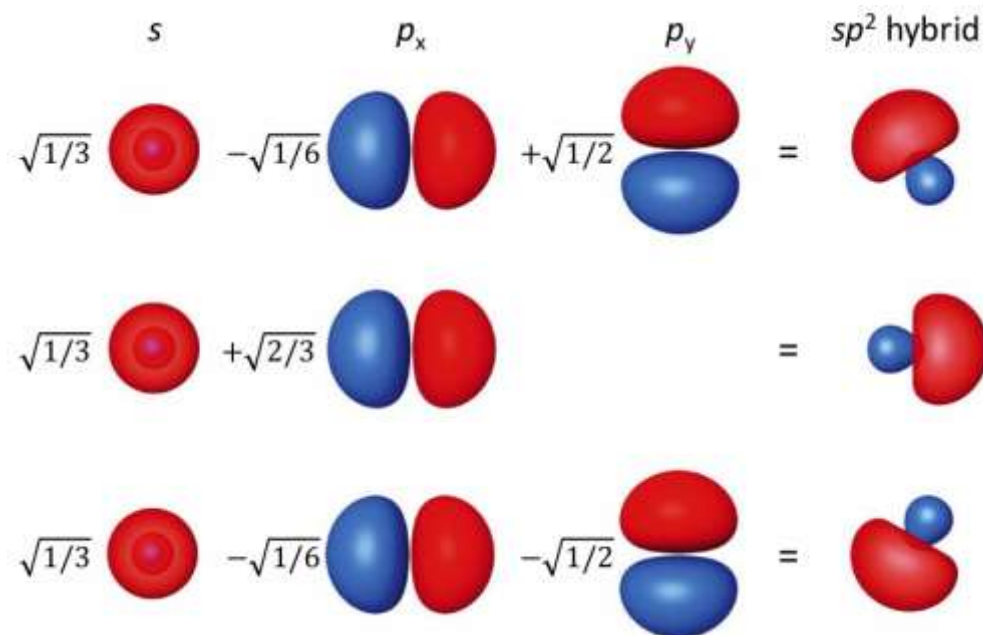
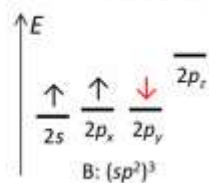
$$\begin{vmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & j & k \end{vmatrix} \begin{vmatrix} s \\ p_x \\ p_y \end{vmatrix}$$



$$\begin{vmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \end{vmatrix} = \begin{vmatrix} 1/\sqrt{3} & \sqrt{2/3} & 0 \\ 1/\sqrt{3} & -1/\sqrt{6} & 1/\sqrt{2} \\ 1/\sqrt{3} & -1/\sqrt{6} & -1/\sqrt{2} \end{vmatrix} \begin{vmatrix} s \\ p_x \\ p_y \end{vmatrix}$$



External field



Hybridization for MO

CH₄ and H₂O (sp³)

sp³ hybrid atomic orbitals: formed from one s orbital and three p orbitals

$$\chi_1(r) = \frac{1}{2} [2s + 2p_x + 2p_y + 2p_z]$$

$$\chi_2(r) = \frac{1}{2} [2s - 2p_x - 2p_y + 2p_z]$$

$$\chi_3(r) = \frac{1}{2} [2s + 2p_x - 2p_y - 2p_z]$$

$$\chi_4(r) = \frac{1}{2} [2s - 2p_x + 2p_y - 2p_z]$$

$$\begin{vmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \\ \varphi_4 \end{vmatrix} = \begin{vmatrix} 1/2 & 1/2 & 1/2 & 1/2 \\ 1/2 & -1/2 & -1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 & -1/2 \\ 1/2 & -1/2 & 1/2 & -1/2 \end{vmatrix} \begin{vmatrix} s \\ p_x \\ p_y \\ p_z \end{vmatrix}$$

